

Lec 1

14th July, 2026

Operations on Signals

1. What is a Signal?

Every signal is defined **with respect to time**.

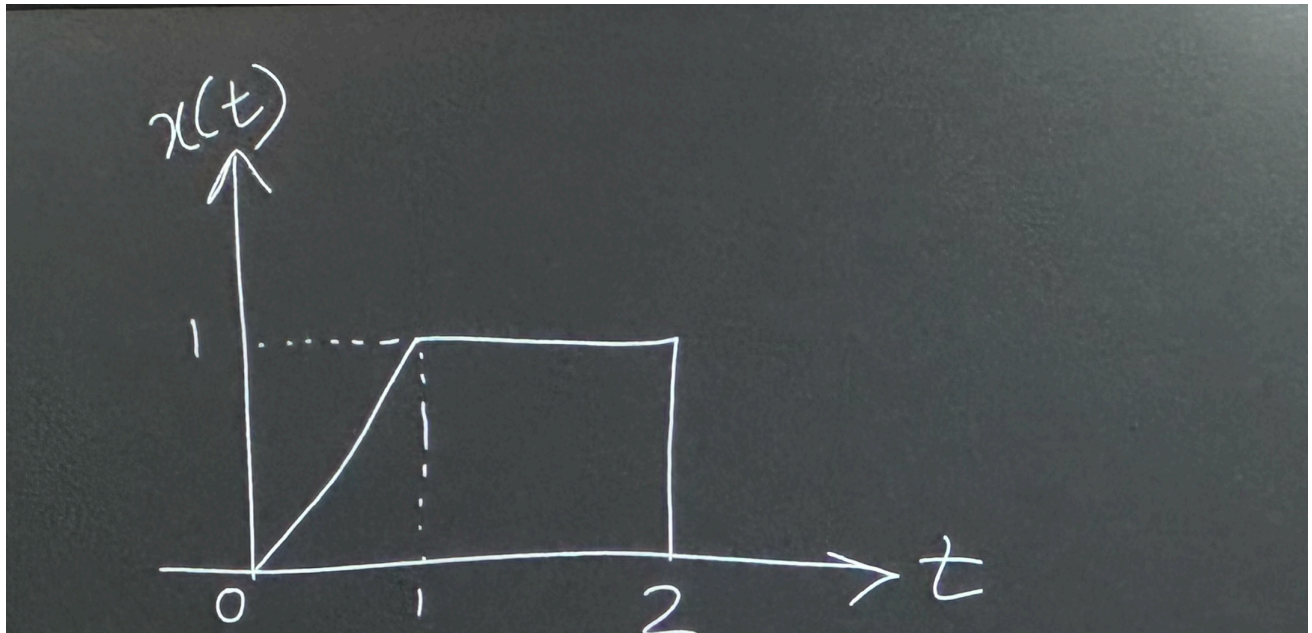
- **Y-axis** → represents value / amplitude
 - **X-axis** → represents time
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2. Types of Signals

Continuous Signal

A signal that has a value at **any given instant of time**.

- Representation: $x(t)$



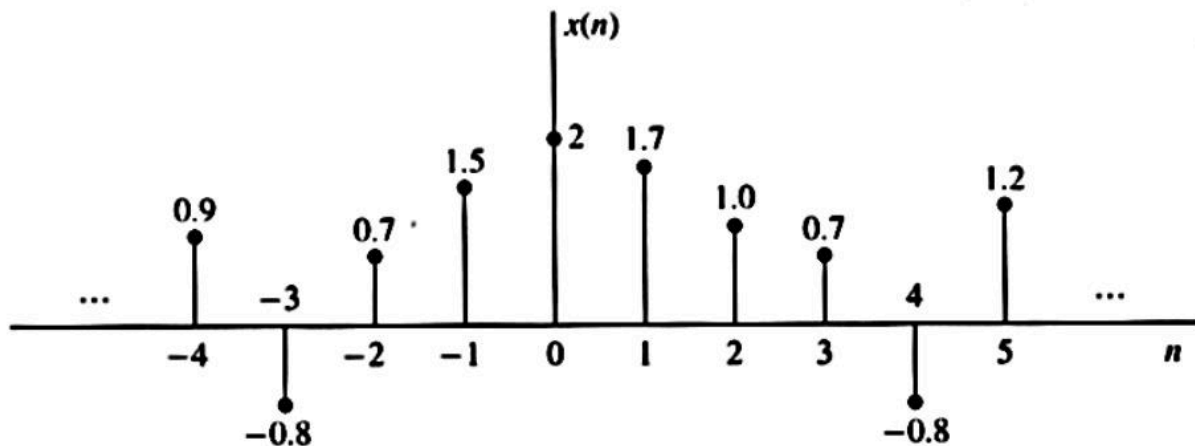
Discrete Signal

A signal that has a value only at **particular instants of time** (not defined in between).

- Representation: $x[n]$

Discrete-Time Signals

- A DTS $x(n)$ is a function of an independent variable that is integer



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3. Scaling Operation

Remember: Whenever a signal undergoes multiplication or division (of the time variable), it is called a **scaling operation**.

In general, for a signal $x(t)$ scaled by a factor a (where $a > 1$):

$$x(at) \rightarrow \text{Compression}$$

$$x\left(\frac{t}{a}\right) \rightarrow \text{Expansion}$$

Two common cases:

Operation	Type	Effect on waveform
$x(2t)$	Multiplication by 2	Compression (signal squeezes toward origin)

Operation	Type	Effect on waveform
$x\left(\frac{t}{2}\right)$	Division by 2	Expansion (signal stretches away from origin)

4. Worked Example

Given: Signal $x(t)$, which starts at $t = 0$ and ends at $t = 2$.

Task: Sketch $x(2t)$ and $x\left(\frac{t}{2}\right)$.

Case 1: $x(2t)$ — Compression



Rule: When the time variable is being *multiplied* by a factor, **divide** the original x-axis values by that same factor to get the new points.

Working:

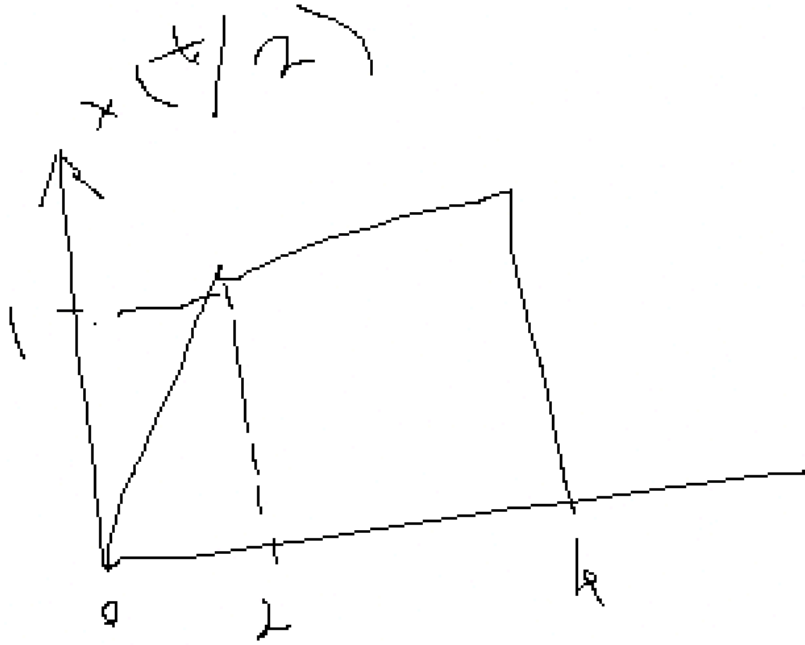
Original points: $t = 0, 1, 2$

New points: $\frac{t}{2} = \frac{0}{2}, \frac{1}{2}, \frac{2}{2} = 0, 0.5, 1$

So the new signal now starts at $t = 0$ and ends at $t = 1$ — it has been **compressed**.

Intuition: Think of watching a YouTube video at 2× speed — the video finishes in half the time. Multiplying the time variable has the same effect: the time axis gets "halved," so the signal duration shrinks.

Case 2: $x\left(\frac{t}{2}\right)$ — Expansion



Rule: When the time variable is being *divided* by a factor, **multiply** the original x-axis values by that same factor to get the new points.

Working:

Original points: $t = 0, 1, 2$

New points: $2t = 0 \times 2, 1 \times 2, 2 \times 2 = 0, 2, 4$

So the new signal now starts at $t = 0$ and ends at $t = 4$ — it has been **expanded**.

5. Summary

- **Continuous signals:** $x(t)$ — defined at every instant
- **Discrete signals:** $x[n]$ — defined only at specific instants
- **Scaling** = multiplying or dividing the time variable
 - $x(at)$ with $a > 1 \rightarrow$ **compresses** the signal (divide x-axis points by a)

- $x\left(\frac{t}{a}\right)$ with $a > 1 \rightarrow$ **expands** the signal (multiply x-axis points by a)